6 Statistics and probability

	Students should be taught to:	Notes
6.1 Graphical representation of data	**A** use different methods of presenting data	Pictograms, bar charts and pie charts, and only two-way tables
	B use appropriate methods of tabulation to enable the construction of statistical diagrams	
	C interpret statistical diagrams	
6.2 Statistical measures	**A** understand the concept of average	Data could be in a list or tabulated form
	B calculate the mean, median, mode and range for a discrete data set	Includes simple problems using these measures
	C calculate an estimate for the mean for grouped data	
	D identify the modal class for grouped data	
6.3 Probability	**A** understand the language of probability	Outcomes, equal likelihood, events, random
	B understand and use the probability scale	P(certainty) = 1 P(impossibility) = 0
	C understand and use estimates or measures of probability from theoretical models	
	D find probabilities from a Venn diagram	
	E understand the concepts of a sample space and an event, and how the probability of an event happening can be determined from the sample space	For the tossing of two coins, the sample space can be listed as: Heads (H), Tails (T): (H, H), (H, T), (T, H), (T, T)
	F list all the outcomes for single events and for two successive events in a systematic way	
	G estimate probabilities from previously collected data	
	H calculate the probability of the complement of an event happening	$P(A') = 1 - P(A)$

	Students should be taught to:	Notes
	I use the addition rule of probability for mutually exclusive events	P(Either A or B occurring) = $\mathrm{P}(A) + \mathrm{P}(B)$ when A and B are mutually exclusive
	J understand and use the term 'expected frequency'	Determine an estimate of the number of times an event with a probability of 0.4 will happen over 300 tries

Higher Tier

Externally assessed

Description

Knowledge of the Foundation Tier content is assumed for students being prepared for the Higher Tier. The Pearson Edexcel International GCSE in Mathematics (Specification A) requires students to demonstrate application and understanding of the following:

Number

- Use numerical skills in a purely mathematical way and in real-life situations.

Algebra

- Use letters as equivalent to numbers and as variables.
- Understand the distinction between expressions, equations and formulae.
- Use algebra to set up and solve problems.
- Demonstrate manipulative skills.
- Construct and use graphs.

Geometry

- Use the properties of angles.
- Understand a range of transformations.
- Work within the metric system.
- Understand ideas of space and shape.
- Use ruler, compasses and protractor appropriately.

Statistics

- Understand basic ideas of statistical averages.
- Use a range of statistical techniques.
- Use basic ideas of probability.

Students should also be able to demonstrate **problem-solving skills** by translating problems in mathematical or non-mathematical contexts into a process or a series of mathematical processes.

Students should be able to demonstrate **mathematical reasoning skills** by:

Assessment information

Each paper is assessed through a 2-hour examination set and marked by Pearson.

The total number of marks for each paper is 100.

Questions will assume knowledge from the Foundation Tier subject content.

Each paper will assess the full range of targeted grades at Higher Tier (9–4).

Each paper will have approximately 40% of the marks distributed evenly over grades 4 and 5 and approximately 60% of the marks distributed evenly over grades 6, 7, 8 and 9.

There will be approximately 40% of questions targeted at grades 5 and 4, across papers 2F and 2H, to aid standardisation and comparability of award between tiers.

Diagrams will not necessarily be drawn to scale and measurements should not be taken from diagrams unless instructions to this effect are given.

Each student may be required to use mathematical instruments, e.g. pair of compasses, ruler, protractor.

A Higher Tier formulae sheet (*Appendix 5*) will be included in the written examinations.

Tracing paper may be used in the examinations.

A calculator may be used in the examinations (please see *page 42* for further information).

Questions will be set in SI units (international system of units).